An Hung Nguyen

COMPUTER SCIENTIST / AI RESEARCHER

Danang, Vietnam

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Google Scholar: An Hung Nguyen

PERSONAL PROFILE

I am a researcher with a strong passion for Artificial Intelligence and Data Science. I have applied AI to real-world problems in medicine and agriculture through collaborations with scientific labs in Vietnam and internationally.

My research interests include Computational Biology, Bioinformatics, Mathematics, Hyper-complex and Quaternion numbers, View Synthesis, Neural Rendering, and Real-Time Graphics.

EDUCATION

University of Science and Technology - The University of Danang *Danang, VN*
B.S., Information Technology *- 2021* 2025

- Grade: 3.65/4.0
- Excellent Bachelor's Degree
- Working as an AI Researcher with Minh Tuan Pham PhD.
- Capstone Project: A Deep Learning-Based Approach for Multi-Weather Image Restoration and Enhancement. (http://thuvienso.dut.udn.vn/handle/DUT/16775)
- Courses: Data Science, Programming for Data Science, Knowledge Representation and Reasoning, Statistical Theory and Methods, Learning Skills through Case Studies, Artificial Intelligence, Machine Learning, Statistical Learning

Le Quy Don upper-secondary school for the gifted *Quang Tri, VN*
Informatics class **(*IT 18-21*)** *- 2018* 2021

- Specialised in Biology, Informatics, and Mathematics with Computer Science
- Selected as a member of the national-level scientific research competition team and the Informatics gifted student team.

WORK EXPERIENCE

SLA Software Logistik Artland GmbH *Germany (Remote)*
Computer Scientist & AI Engineer *- Aug 2024* Sept 2026

- Support of integration of AI models into existing customer infrastructure.
- Collaborated with an AI team to develop neural network architectures to detect and segment animal pathogens.
- Developed scripts and applications to automate and handle data.
- Automated and optimized the data handling process for signal processing, worked with Linux distributions, performed shell scripting, and utilized various Linux tools.
- Technical Skills: Python with PyTorch, NumPy, Matplotlib, Pandas, Scikit-learn, C++, Ubuntu Linux, Linux tools, Apt, Scripting, Git.
- Soft Skills: Teamwork, Time Management, Communication, Presentation skills.
- Research Skills: Medical Research, Data Analysis, Data Engineering.

University of Regina *Canada*
Collaboration Researcher *- Aug 2024* Sept 2025

- Researched solutions for simulating chemical processes.
- Implemented simulation code using Matlab and Python.

Phygital Labs *Vietnam*
AI Intern *- May 2024* Jul 2024

- Worked as an AI intern on deep learning projects.
- Worked with LLM models to propose solutions for querying large documents.
- Collaborated with a former Senior Google engineer to deploy systems.

FX Data *Germany (Remote)*
Computer Scientist & AI Engineer *- Jan 2022* Jul 2024

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- Soft Skills: Teamwork, Time Management, Communication, Presentation skills.
- Research Skills: Medical Research, Data Analysis, Data Engineering.

Rhen Consulting *Germany (Remote)*
AI Engineer *- Oct 2021* Dec 2021

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- Developed scripts and applications to automate and handle data.
- Automated and optimized the data handling process for signal processing, worked with Linux distributions, performed shell scripting, and utilized various Linux tools.

HIGHLIGHT AWARDS

Fourth Place Award in the Field of Science, Engineering, and Technology *Vietnam*
Vietnam Student Science Research and Innovation 2022 - 2023 Dec 2023
Project title: Quaternion Convolutional Neural Networks for Image Depth Estimation

Fourth Place Award in the in Computational Biology and Bioinformatics *Vietnam*
Vietnam National Secondary Student Science and Engineering Fair March 2021
Project title: Applying Deep Learning to the process of diagnosing, analyzing and detecting damage to the retina due to diabetes

Third Prize in the Provincial Contest for Excellent Students in Informatics *Vietnam*
Quang Tri Department of Education and Training Jan 2020

PUBLICATIONS

[1] **All-in-One Image Restoration with Quaternion Cross-Attention and Haar Frequency Decomposition**
Submitted to Signal, Image and Video Processing (Journal Q2) 2025
Status: Under review
An Hung Nguyen, Minh Tuan Pham
dx.doi.org/10.21203/rs.3.rs-7260641/v1
This work is part of my undergraduate research project and capstone thesis.

[2] **Optimizing Retrieval in Vietnamese Literary QA Using Named-Entity Relationship Graphs**
Submitted to ACM Transactions on Asian and Low-Resource Language Information Processing (Q2) 2025
Status: Under review
Minh Tuan Pham, **An Hung Nguyen**, Phuc Hao Do, Thanh Huy Nguyen, and Nam Do

[3] **Real-Time 3D Reconstruction through Image Synthesis and IoT Integration**
Manuscript under review 2025

Cong Thang PHAM, **An Hung Nguyen**, Tan Sang Truong, Dai Quy Tran, Si Tien Tran
Submitted to Vestnik St. Petersburg University: Mathematics

[4] **Adaptive Hybrid Transformer Network with Frequency Attention for All-in-One Image Restoration**
PSBB25 - Moscow, RUSSIA - June 09 11, 2025

Cong Thang Pham, **An Hung Nguyen**, Quoc Cuong Nguyen, Minh Nhat Phan, Thanh Than Nguyen
10.5194/isprs-archives-XLVIII-2-W9-2025-213-2025

[5] **DehazeCLNet: A Contrastive Learning Framework with Advanced Feature Extraction for Image Dehazing**
SOICT 2024 (Springer CCIS) - Danang, Vietnam December 13-15, 2024

Cong Thang Pham, **An Hung Nguyen**, Quoc Cuong Nguyen, Minh Nhat Phan
10.1007/978-981-96-4282-3_23

[6] **Enhancing Image Classification Capabilities In The Vision Transformer Network Model With Quaternion Algebra**
CITA 2024 - Danang, Vietnam July, 2024

Minh Tuan Pham, **An Hung Nguyen**
https://elib.vku.udn.vn/handle/123456789/4031
https://nguyenhung.me/paper/QViT.pdf

[7] **Enhancing Convolutional Neural Network Architectures with Long Short-Term Memory for Improved Image Classification**
IEEE: Institute of Electrical and Electronics Engineers - Danang, Vietnam November 10-11, 2023

An Hung Nguyen, Minh Tuan Pham
10.1109/ATiGB59969.2023.10364368

[8] **Quaternion Convolutional Neural Networks for Depth Estimation**
IEEE: Institute of Electrical and Electronics Engineers - Mie, Japan January 4, 2023

An Hung Nguyen, Cao Duy Hoang, Dang Hoang Phu Phan, Minh Tuan Pham
https://doi.org/10.1109/SCISISIS55246.2022.10002074

[9] **Weight Initialization for Complex Number Convolutional Neural Networks Using Genetic Algorithms**
Journal on Information Technologies and Communications - Vietnam June 30, 2022

Minh Tuan Pham, **An Hung Nguyen**
https://doi.org/10.32913/mic-ict-research-vn.v2022.n1.1054

[10] **Efficient Complex Valued Neural Network with Fourier Transform on Image Denoising**
ACM: Association for Computing Machinery - Dubai, UAE December 16, 2021

Minh Tuan Pham, Van Quang Nguyen, Cao Duy Hoang, **An Hung Nguyen**
https://doi.org/10.1145/3508072.3508081

SKILLS

Programming:	C/C++, Python (Pandas, PyTorch, NumPy, Scikit-learn. etc), Java, dot NET, React Native, SQL.
Miscellaneous:	Linux, Shell (Bash/Zsh), LATEX(Overleaf/R Markdown), Microsoft Office, Git, Arduino, MCU microcontroller.
Soft Skills:	Time Management, Teamwork, Leadership, Problem-solving, Documentation, Engaging Presentation.

TOP HONORS AND AWARDS

Mar 2025	2nd prizes, Faculty level Student Scientific Research and Innovation - Vietnam
Dec 2023	Third prize, University-level Student Scientific Research and Innovation - Vietnam
Dec 2023	First prize, Faculty level Student Scientific Research and Innovation - Vietnam
Dec 2023	4th prizes, Vietnam Student Science Research and Innovation - Vietnam
Jun 2023	3rd prizes, Faculty level Student Scientific Research and Innovation - Vietnam
Mar 2022	2nd prizes, Faculty level Student Science Research and Innovation - Vietnam
Aug 2021	Second prize (ranked 1st), Innovative Generation 2021 - Vietnam
Mar 2021	4th prizes, Vietnam National Secondary Student Science and Engineering Fair - Vietnam
Jan 2021	4th prizes, Provincial Contest for Excellent Students in Informatics - Vietnam
Jan 2020	3rd prizes, Provincial Contest for Excellent Students in Informatics - Vietnam

LANGUAGES

English: Professional Proficiency (B2)

Chinese: Minimum Professional Proficiency

Vietnamese: Native proficiency